

Assignment for Lecture 6

1. Implementation of Student Class

a) Write a class Student which models students with their:

ID: string, name, accumulated CPA, prefer courses (an array of String).

b) Write a program that create and display about the following students:

(TROY1231, Bob Walker, 3.21, {"Math", "Philosophy", "Art"})

(TROY1217, Bao Trung, 2.67, {"Computer Science I", "Math"})

(TROY1362, Nina Gigi, 2.99, {"Sing", "Dance", "Physics"})

the find the course that is most prefer by Student.

Don't use constructor, private methods in this exercises.

2. Implementation of TV Class

Consider television sets. Each TV is an object with states (current channel, current volume level, power on or off) and behaviors (change channels, adjust volume, turn on/off). Write a class to model TV sets and demo their usages in a program.

TV	
channel: int volumeLevel: int on: boolean	The current channel (1 to 120) of this TV. The current volume level (1 to 7) of this TV. Indicates whether this TV is on/off.
+TV() +turnOn(): void +turnOff(): void +setChannel(newChannel: int): void +setVolume(newVolumeLevel: int): void +channelUp(): void +channelDown(): void +volumeUp(): void +volumeDown(): void	Constructs a default TV object. Turns on this TV. Turns off this TV. Sets a new channel for this TV. Sets a new volume level for this TV. Increases the channel number by 1. Decreases the channel number by 1. Increases the volume level by 1. Decreases the volume level by 1.

3. Programming Challenges (Array topic)

6.1, 6.2, 6.3, 6.4 Tony Gaddis Java